

# Non-Linear Modeling

## Chapter 7 lab handout

Last modified: 2026-09-29

This handout adapts the Chapter 7 Python lab from the *Introduction to Statistical Learning with Applications in Python* (ISLP) textbook, with permission from the source available at <https://www.statlearning.com>. The original lab is `Ch07-nonlin-lab.Rmd` in the `intro-stat-learning/ISLP_labs` repository.

Students who prefer **R** can work through the corresponding chapter of Emil Hvitfeldt's [ISLR tidymodels labs](#).

In this lab we demonstrate some of the nonlinear models discussed in this chapter. We use the `Wage` data as a running example, and show that many of the complex non-linear fitting procedures discussed can easily be implemented in Python.

## 1 Setup

As usual, we start with some of our standard imports.

```
import numpy as np, pandas as pd
from matplotlib.pyplot import subplots
import statsmodels.api as sm
from ISLP import load_data
from ISLP.models import (summarize,
                        poly,
                        ModelSpec as MS)
from statsmodels.stats.anova import anova_lm
```

We again collect the new imports needed for this lab. Many of these are developed specifically for the ISLP package.

```
from pygam import (s as s_gam,
                  l as l_gam,
                  f as f_gam,
                  LinearGAM,
                  LogisticGAM)

from ISLP.transforms import (BSpline,
                            NaturalSpline)
```

```
from ISLP.models import bs, ns
from ISLP.pygam import (approx_lam,
                        degrees_of_freedom,
                        plot as plot_gam,
                        anova as anova_gam)
```

## 2 Polynomial Regression and Step Functions

**Exercise 1.** Load the Wage data with `load_data('Wage')`. Store the response (wage) as `y` and the predictor (age) as `age`.

Solution

```
Wage = load_data('Wage')
y = Wage['wage']
age = Wage['age']
```

Throughout most of this lab, our response is `Wage['wage']`, which we have stored as `y` above.

**Exercise 2.** Use `poly()` inside `MS()` to build a 4th-degree polynomial in `age`, fit an OLS model of `y` on this basis, and print the coefficient summary.

Solution

```
poly_age = MS([poly('age', degree=4)]).fit(Wage)
M = sm.OLS(y, poly_age.transform(Wage)).fit()
summarize(M)
```

|                        | coef      | std err | t       | P> t  |
|------------------------|-----------|---------|---------|-------|
| intercept              | 111.7036  | 0.729   | 153.283 | 0.000 |
| poly(age, degree=4)[0] | 447.0679  | 39.915  | 11.201  | 0.000 |
| poly(age, degree=4)[1] | -478.3158 | 39.915  | -11.983 | 0.000 |
| poly(age, degree=4)[2] | 125.5217  | 39.915  | 3.145   | 0.002 |
| poly(age, degree=4)[3] | -77.9112  | 39.915  | -1.952  | 0.051 |

`poly()` is a *helper* that sets up the transformation; `Poly()` is the workhorse that computes it. The first line above runs the `fit()` method on `Wage`, which stores any parameters needed by `Poly()` on the training data so that subsequent `transform()` calls (on new data, or inside the plotting function below) use the same basis.

**Exercise 3.** Build a grid `age_grid` of 100 evenly spaced values spanning the observed range of `age`, and wrap it in a single-column `DataFrame` named `age_df`.

Solution

```
age_grid = np.linspace(age.min(),
                        age.max(),
                        100)
age_df = pd.DataFrame({'age': age_grid})
```

**Exercise 4.** Write a function `plot_wage_fit(age_df, basis, title)` that fits OLS on the supplied basis, evaluates it on `age_df`, and plots the raw data with the fitted curve and its 95% confidence bands.

Solution

```
def plot_wage_fit(age_df,
                  basis,
                  title):

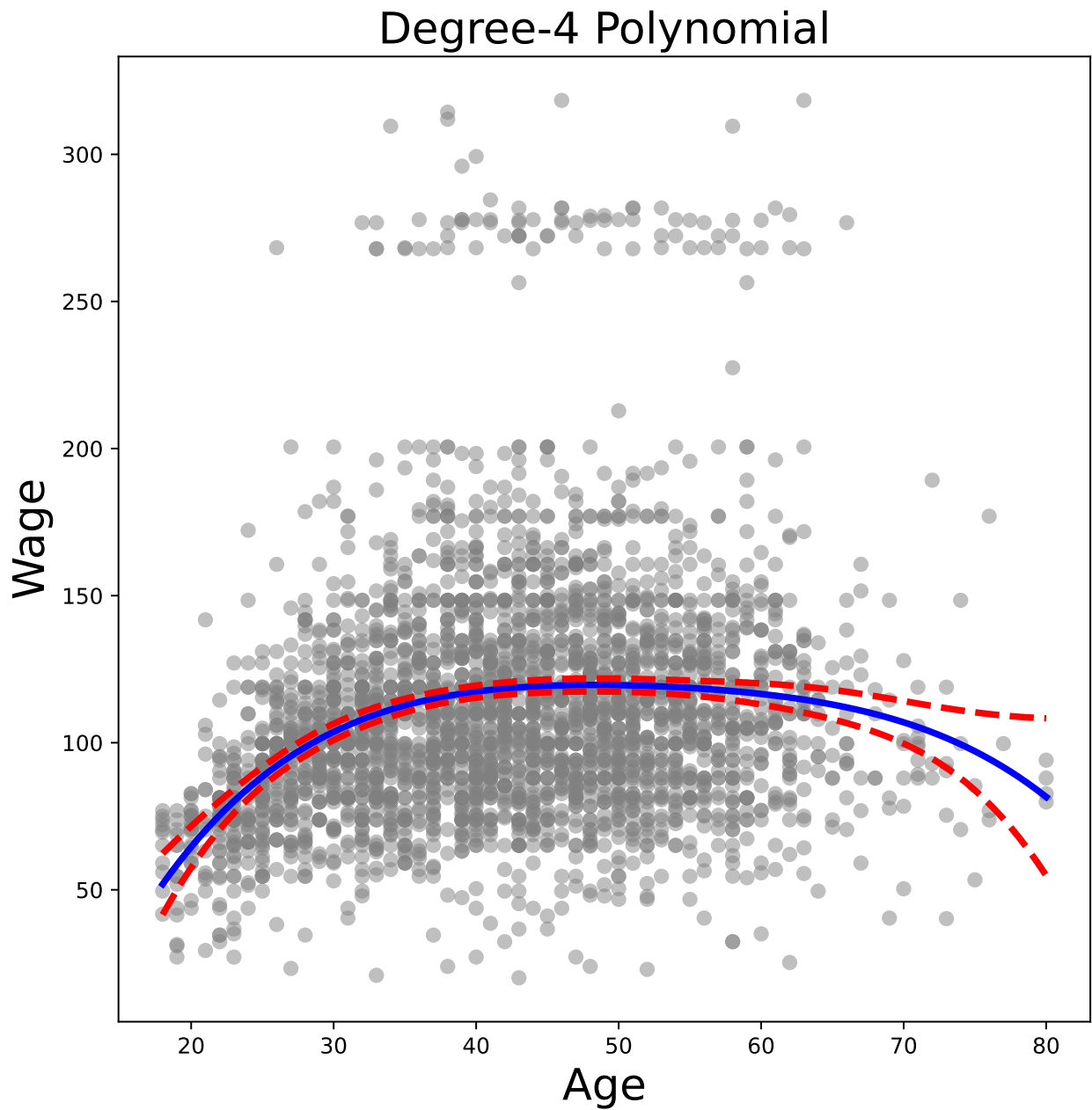
    X = basis.transform(Wage)
    Xnew = basis.transform(age_df)
    M = sm.OLS(y, X).fit()
    preds = M.get_prediction(Xnew)
    bands = preds.conf_int(alpha=0.05)
    fig, ax = subplots(figsize=(8, 8))
    ax.scatter(age,
               y,
               facecolor='gray',
               alpha=0.5)
    for val, ls in zip([preds.predicted_mean,
                       bands[:, 0],
                       bands[:, 1]],
                       ['b', 'r--', 'r--']):
        ax.plot(age_df.values, val, ls, linewidth=3)
    ax.set_title(title, fontsize=20)
    ax.set_xlabel('Age', fontsize=20)
    ax.set_ylabel('Wage', fontsize=20)
    return ax
```

The `alpha` argument to `ax.scatter()` adds transparency that gives a visual indication of density. `zip()` bundles the three lines (fitted mean and two confidence bands) with their colours and line types for the `for` loop.

**Exercise 5.** Use your helper to plot the degree-4 polynomial fit.

Solution

```
plot_wage_fit(age_df,
              poly_age,
              'Degree-4 Polynomial');
```



**Exercise 6.** With polynomial regression we must decide on the degree of the polynomial to use. Fit polynomial models in `age` of degrees 1 through 5, and use `anova_lm()` to test which degree is sufficient.

Solution

```
models = [MS([poly('age', degree=d)])  
           for d in range(1, 6)]  
Xs = [model.fit_transform(Wage) for model in models]  
anova_lm([sm.OLS(y, X_).fit()  
          for X_ in Xs])
```

|   | df_resid | ssr          | df_diff | ss_diff       | F          | Pr(>F)       |
|---|----------|--------------|---------|---------------|------------|--------------|
| 0 | 2998.0   | 5.022216e+06 | 0.0     | NaN           | NaN        | NaN          |
| 1 | 2997.0   | 4.793430e+06 | 1.0     | 228786.010128 | 143.593107 | 2.363850e-32 |
| 2 | 2996.0   | 4.777674e+06 | 1.0     | 15755.693664  | 9.888756   | 1.679202e-03 |
| 3 | 2995.0   | 4.771604e+06 | 1.0     | 6070.152124   | 3.809813   | 5.104620e-02 |
| 4 | 2994.0   | 4.770322e+06 | 1.0     | 1282.563017   | 0.804976   | 3.696820e-01 |

The `*` unpacks the list of fitted models for `anova_lm()`'s variable argument list. The p-value comparing linear `models[0]` to quadratic `models[1]` is essentially zero, and the cubic-vs-quartic comparison is roughly 5%, while degree 5 is unnecessary ( $p = 0.37$ ). Either a cubic or a quartic polynomial gives a reasonable fit.

**Exercise 7.** Repeat the ANOVA, but with a linear term in `education` included in each model alongside a polynomial in `age` of degrees 1, 2, 3.

Solution

```
models = [MS(['education', poly('age', degree=d)])
           for d in range(1, 4)]
XEs = [model.fit_transform(Wage)
        for model in models]
anova_lm(*[sm.OLS(y, X_).fit() for X_ in XEs])
```

|   | df_resid | ssr          | df_diff | ss_diff       | F          | Pr(>F)       |
|---|----------|--------------|---------|---------------|------------|--------------|
| 0 | 2997.0   | 3.902335e+06 | 0.0     | NaN           | NaN        | NaN          |
| 1 | 2996.0   | 3.759472e+06 | 1.0     | 142862.701185 | 113.991883 | 3.838075e-26 |
| 2 | 2995.0   | 3.753546e+06 | 1.0     | 5926.207070   | 4.728593   | 2.974318e-02 |

The ANOVA approach works whether or not the polynomials are orthogonal, provided the models are *nested*. As an alternative, we could choose the polynomial degree by cross-validation (Chapter 5).

**Exercise 8.** Now consider predicting whether someone earns more than \$250,000 per year. Fit a logistic regression (use `sm.GLM` with a `Binomial` family) of `wage > 250` on the same 4th-degree polynomial basis in `age`, and produce a plot of the predicted probability vs. `age` (with confidence bands and a rug plot).

Solution

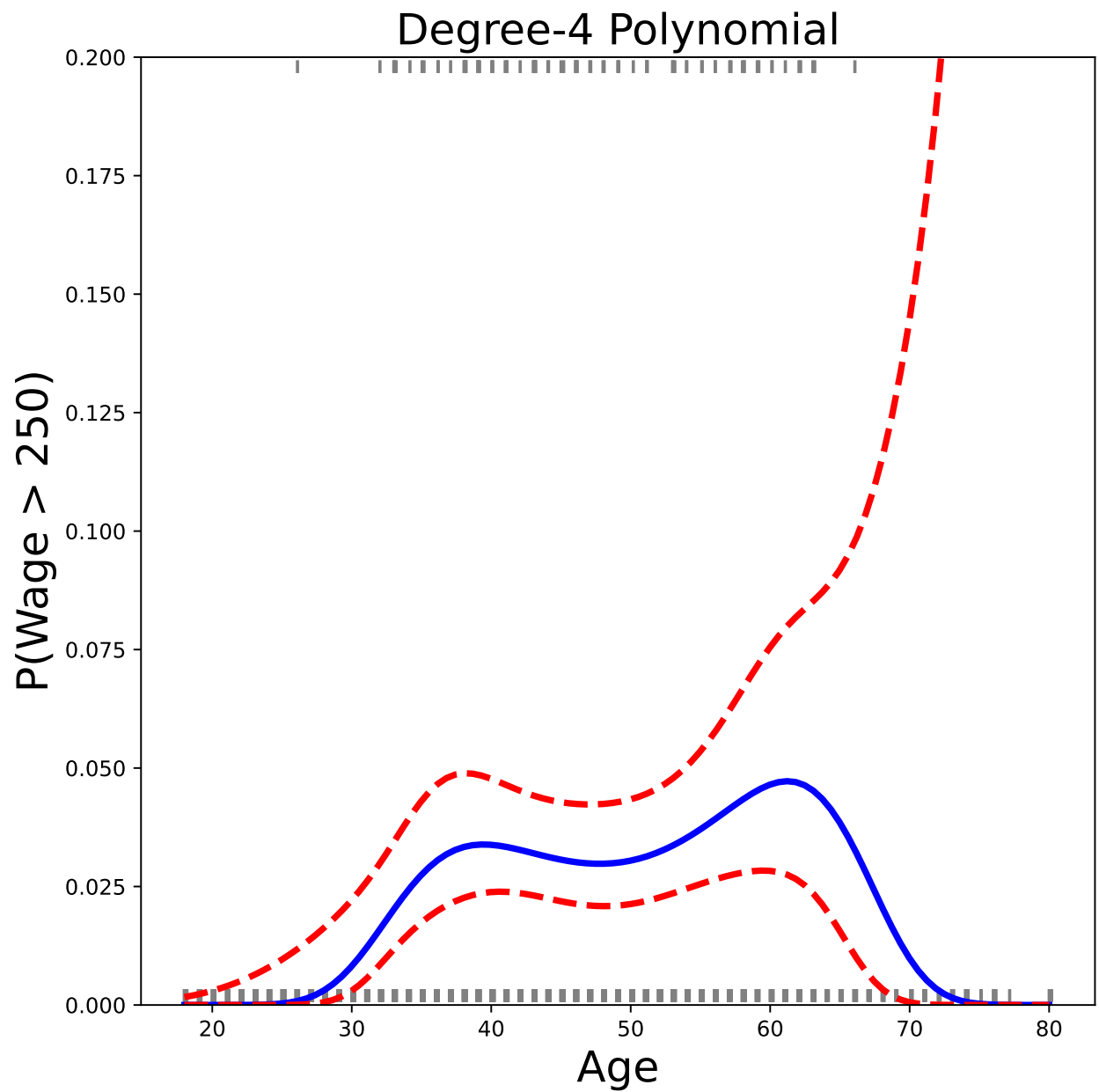
```
X = poly_age.transform(Wage)
high_earn = Wage['high_earn'] = y > 250 # shorthand
glm = sm.GLM(y > 250,
             X,
             family=sm.families.Binomial())
B = glm.fit()
summarize(B)
```

|                        | coef     | std err | z       | P> z  |
|------------------------|----------|---------|---------|-------|
| intercept              | -4.3012  | 0.345   | -12.457 | 0.000 |
| poly(age, degree=4)[0] | 71.9642  | 26.133  | 2.754   | 0.006 |
| poly(age, degree=4)[1] | -85.7729 | 35.929  | -2.387  | 0.017 |
| poly(age, degree=4)[2] | 34.1626  | 19.697  | 1.734   | 0.083 |
| poly(age, degree=4)[3] | -47.4008 | 24.105  | -1.966  | 0.049 |

(The double assignment `high_earn = Wage['high_earn'] = y > 250` both binds the local variable `high_earn` and stores the indicator as a new column on the `Wage` DataFrame. We'll re-use that column in Exercises 29 and 30.)

```
newX = poly_age.transform(age_df)
preds = B.get_prediction(newX)
bands = preds.conf_int(alpha=0.05)

fig, ax = subplots(figsize=(8, 8))
rng = np.random.default_rng(0)
ax.scatter(age +
            0.2 * rng.uniform(size=y.shape[0]),
            np.where(high_earn, 0.198, 0.002),
            fc='gray',
            marker='|')
for val, ls in zip([preds.predicted_mean,
                    bands[:, 0],
                    bands[:, 1]],
                    ['b', 'r--', 'r--']):
    ax.plot(age_df.values, val, ls, linewidth=3)
ax.set_title('Degree-4 Polynomial', fontsize=20)
ax.set_xlabel('Age', fontsize=20)
ax.set_ylim([0, 0.2])
ax.set_ylabel('P(Wage > 250)', fontsize=20);
```



The `age` values above 250 are shown as gray marks at the top of the plot; those below 250 at the bottom. A small amount of jitter prevents identical-age points from overlapping. This is called a *rug plot*.

**Exercise 9.** Fit a step-function regression: discretise `age` into 4 quantile-based bins with `pd.qcut()`, build dummy columns with `pd.get_dummies()`, and fit OLS.

Solution

```
cut_age = pd.qcut(age, 4)
summarize(sm.OLS(y, pd.get_dummies(cut_age)).fit())
```

|                 | coef     | std err | t      | P> t |
|-----------------|----------|---------|--------|------|
| (17.999, 33.75] | 94.1584  | 1.478   | 63.692 | 0.0  |
| (33.75, 42.0]   | 116.6608 | 1.470   | 79.385 | 0.0  |
| (42.0, 51.0]    | 119.1887 | 1.416   | 84.147 | 0.0  |
| (51.0, 80.0]    | 116.5717 | 1.559   | 74.751 | 0.0  |

`pd.qcut()` chose cutpoints at the 25%, 50% and 75% quantiles, so we get four bins. `pd.get_dummies()` (unlike the usual model-matrix convention) keeps *all* category levels; combined with no intercept, each coefficient is the average wage in its bin. For non-quantile cutpoints, use `pd.cut()` instead.

### 3 Splines

**Exercise 10.** Use `BSpline()` from `ISLP.transforms` with `internal_knots=[25, 40, 60]` and `intercept=True` to build a cubic B-spline basis for `age`. What shape is the resulting design matrix?

Solution

```
bs_ = BSpline(internal_knots=[25, 40, 60], intercept=True).fit(age)
bs_age = bs_.transform(age)
bs_age.shape
```

(3000, 7)

A cubic B-spline with 3 interior knots yields a 7-column matrix (3 + degree + 1 with `intercept=True`).

**Exercise 11.** Now fit a cubic spline regression of `wage` on `age` using the helper `bs()` inside `MS()`. Pass `name='bs(age)'` to clean up the column labels in the summary.

Solution

```
bs_age = MS([bs('age',
               internal_knots=[25, 40, 60],
               name='bs(age)')])
Xbs = bs_age.fit_transform(Wage)
M = sm.OLS(y, Xbs).fit()
summarize(M)
```

|            | coef    | std err | t     | P> t  |
|------------|---------|---------|-------|-------|
| intercept  | 60.4937 | 9.460   | 6.394 | 0.000 |
| bs(age)[0] | 3.9805  | 12.538  | 0.317 | 0.751 |
| bs(age)[1] | 44.6310 | 9.626   | 4.636 | 0.000 |
| bs(age)[2] | 62.8388 | 10.755  | 5.843 | 0.000 |
| bs(age)[3] | 55.9908 | 10.706  | 5.230 | 0.000 |

```
bs(age)[4] 50.6881 14.402 3.520 0.000
bs(age)[5] 16.6061 19.126 0.868 0.385
```

Note there are 6 spline coefficients rather than 7 because `bs()` defaults to `intercept=False` — the basis would normally be 7 columns, but one is dropped to accommodate the model’s overall intercept.

**Exercise 12.** Instead of specifying knots, ask `BSpline` for `df=6`. Inspect `internal_knots_` to see where the knots ended up.

Solution

```
BSpline(df=6).fit(age).internal_knots_
```

```
array([33.75, 42.    , 51.   ])
```

With `df=6` and cubic splines, three knots are placed at the 25%, 50% and 75% quantiles of `age` — ages 33.75, 42.0, and 51.0.

**Exercise 13.** B-splines need not be cubic. Fit a degree-0 spline with `df=3` and compare the coefficients to your earlier `qcut()` step-function fit.

Solution

```
bs_age0 = MS([bs('age',
                 df=3,
                 degree=0)]).fit(Wage)
Xbs0 = bs_age0.transform(Wage)
summarize(sm.OLS(y, Xbs0).fit())
```

|                            | coef    | std err | t      | P> t |
|----------------------------|---------|---------|--------|------|
| intercept                  | 94.1584 | 1.478   | 63.687 | 0.0  |
| bs(age, df=3, degree=0)[0] | 22.3490 | 2.152   | 10.388 | 0.0  |
| bs(age, df=3, degree=0)[1] | 24.8076 | 2.044   | 12.137 | 0.0  |
| bs(age, df=3, degree=0)[2] | 22.7814 | 2.087   | 10.917 | 0.0  |

Degree-0 B-splines are piecewise constant — equivalent to the step function from Exercise 9. With `df=3` (degree 0), the knots are again at the same three quantiles. The coefficients look different because of the coding: here the intercept is a column of ones and the remaining three coefficients are *increments* relative to the first bin. The sums approximately recover the per-bin means (slight differences are because `qcut()` uses `while` while `bs()` uses `<` at the boundaries).

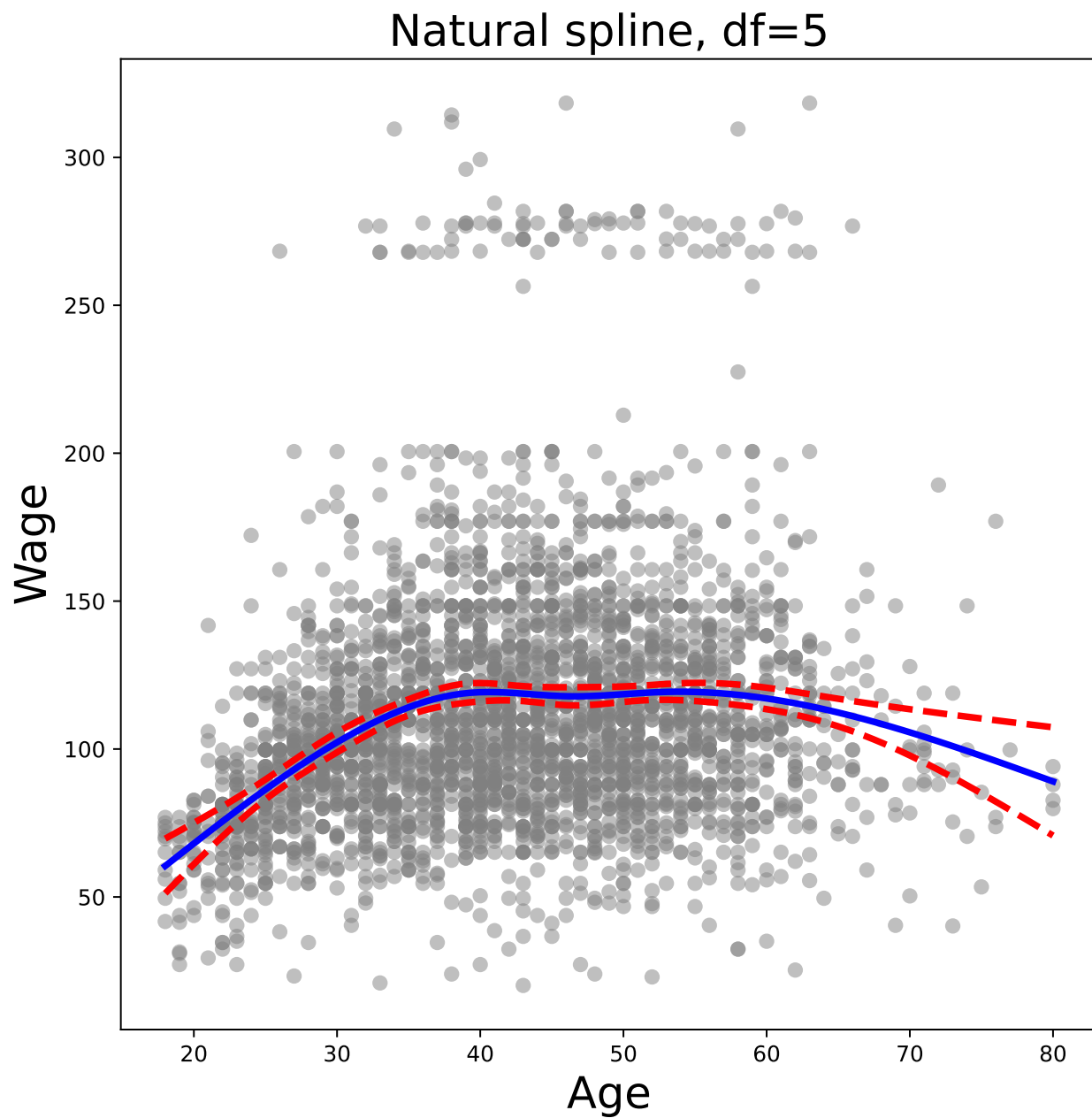
**Exercise 14.** Fit a natural cubic spline with 5 degrees of freedom using `ns()`, and plot the fit with `plot_wage_fit()`.

## Solution

```
ns_age = MS([ns('age', df=5)]).fit(Wage)
M_ns = sm.OLS(y, ns_age.transform(Wage)).fit()
summarize(M_ns)
```

|                  | coef    | std err | t      | P> t  |
|------------------|---------|---------|--------|-------|
| intercept        | 60.4752 | 4.708   | 12.844 | 0.000 |
| ns(age, df=5)[0] | 61.5267 | 4.709   | 13.065 | 0.000 |
| ns(age, df=5)[1] | 55.6912 | 5.717   | 9.741  | 0.000 |
| ns(age, df=5)[2] | 46.8184 | 4.948   | 9.463  | 0.000 |
| ns(age, df=5)[3] | 83.2036 | 11.918  | 6.982  | 0.000 |
| ns(age, df=5)[4] | 6.8770  | 9.484   | 0.725  | 0.468 |

```
plot_wage_fit(age_df,  
              ns_age,  
              'Natural spline, df=5');
```



## 4 Smoothing Splines and GAMs

**Exercise 15.** A smoothing spline is a special case of a GAM with squared-error loss and a single feature. Fit `LinearGAM(s_gam(0, lam=0.6))` to predict wage from age. (Remember `pygam` expects a 2-D feature matrix.)

### Solution

```
X_age = np.asarray(age).reshape((-1, 1))
gam = LinearGAM(s_gam(0, lam=0.6))
gam.fit(X_age, y)
```

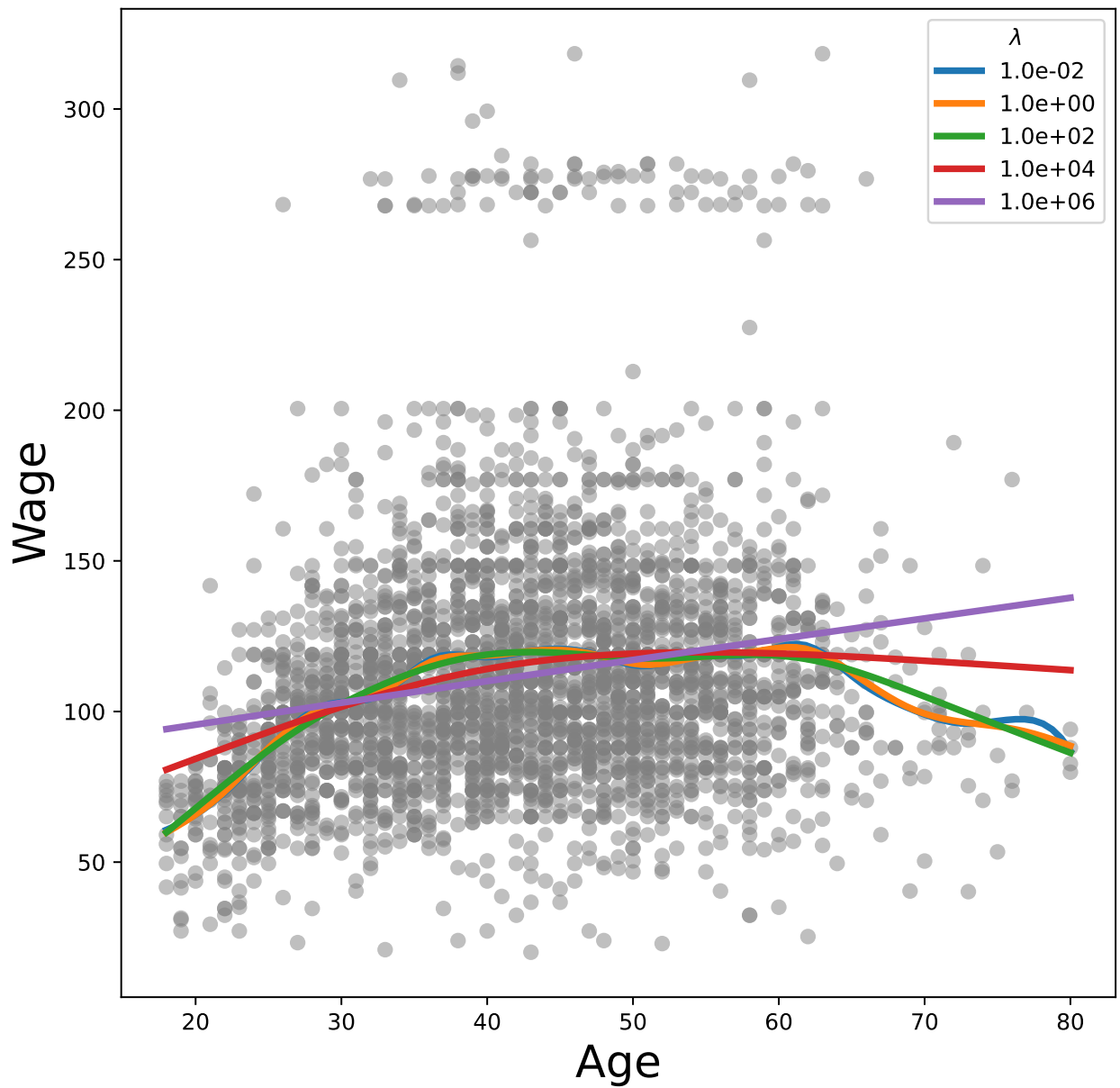
```
LinearGAM(callbacks=[Deviance(), Diffs()], fit_intercept=True,
           max_iter=100, scale=None, terms=s(0) + intercept, tol=0.0001,
           verbose=False)
```

`s_gam(0, ...)` says “apply a smoothing spline to column 0 of the feature matrix”. `lam` is the penalty parameter  $\lambda$ . The `-1` inside `.reshape((-1, 1))` tells NumPy to infer that dimension’s size.

**Exercise 16.** Investigate how the fit changes with the smoothing parameter. Loop `lam` over `np.logspace(-2, 6, 5)` and plot each resulting fit.

### Solution

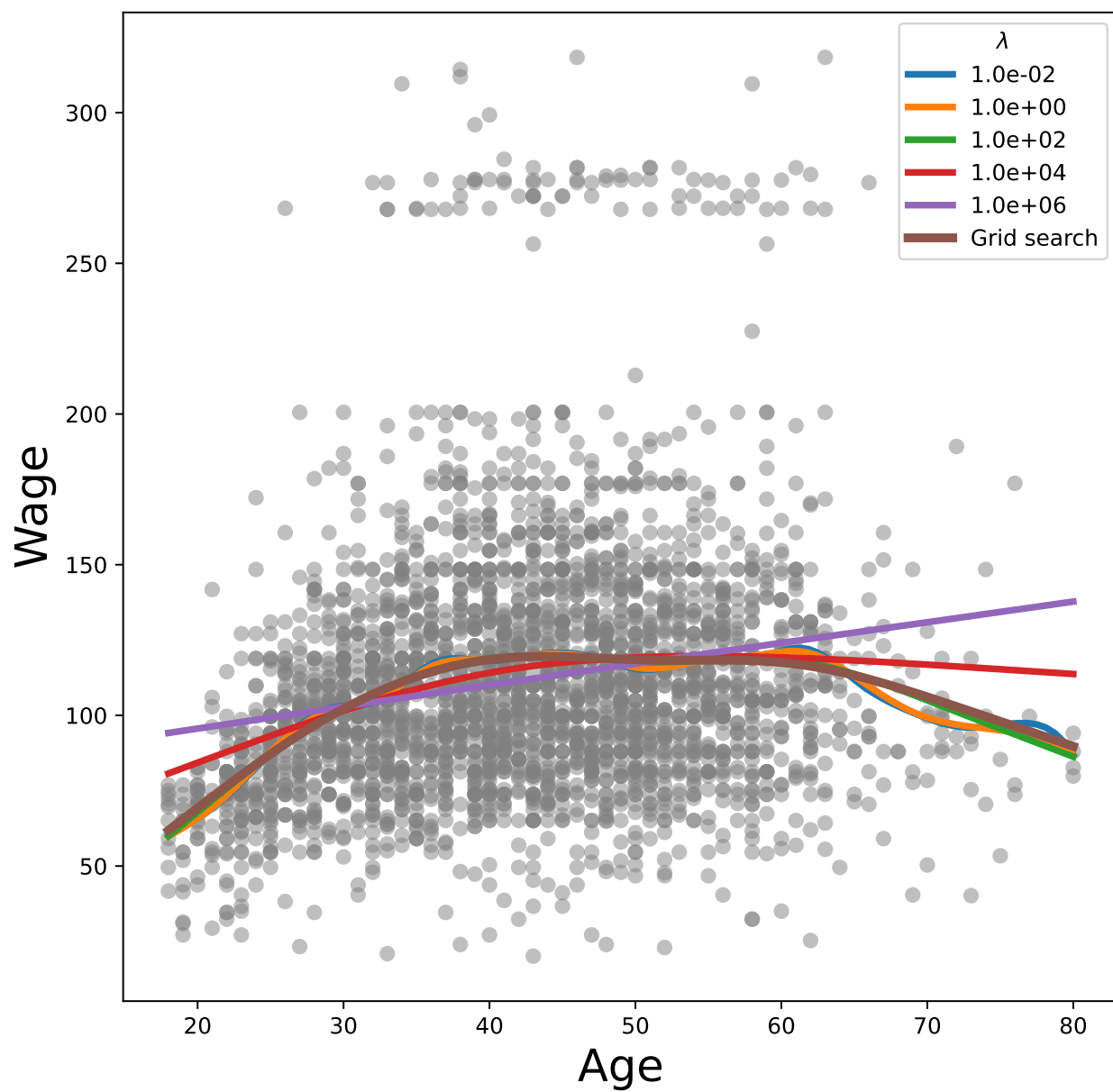
```
fig, ax = subplots(figsize=(8, 8))
ax.scatter(age, y, facecolor='gray', alpha=0.5)
for lam in np.logspace(-2, 6, 5):
    gam = LinearGAM(s_gam(0, lam=lam)).fit(X_age, y)
    ax.plot(age_grid,
            gam.predict(age_grid),
            label='{:.1e}'.format(lam),
            linewidth=3)
ax.set_xlabel('Age', fontsize=20)
ax.set_ylabel('Wage', fontsize=20);
ax.legend(title=r'$\lambda$');
```

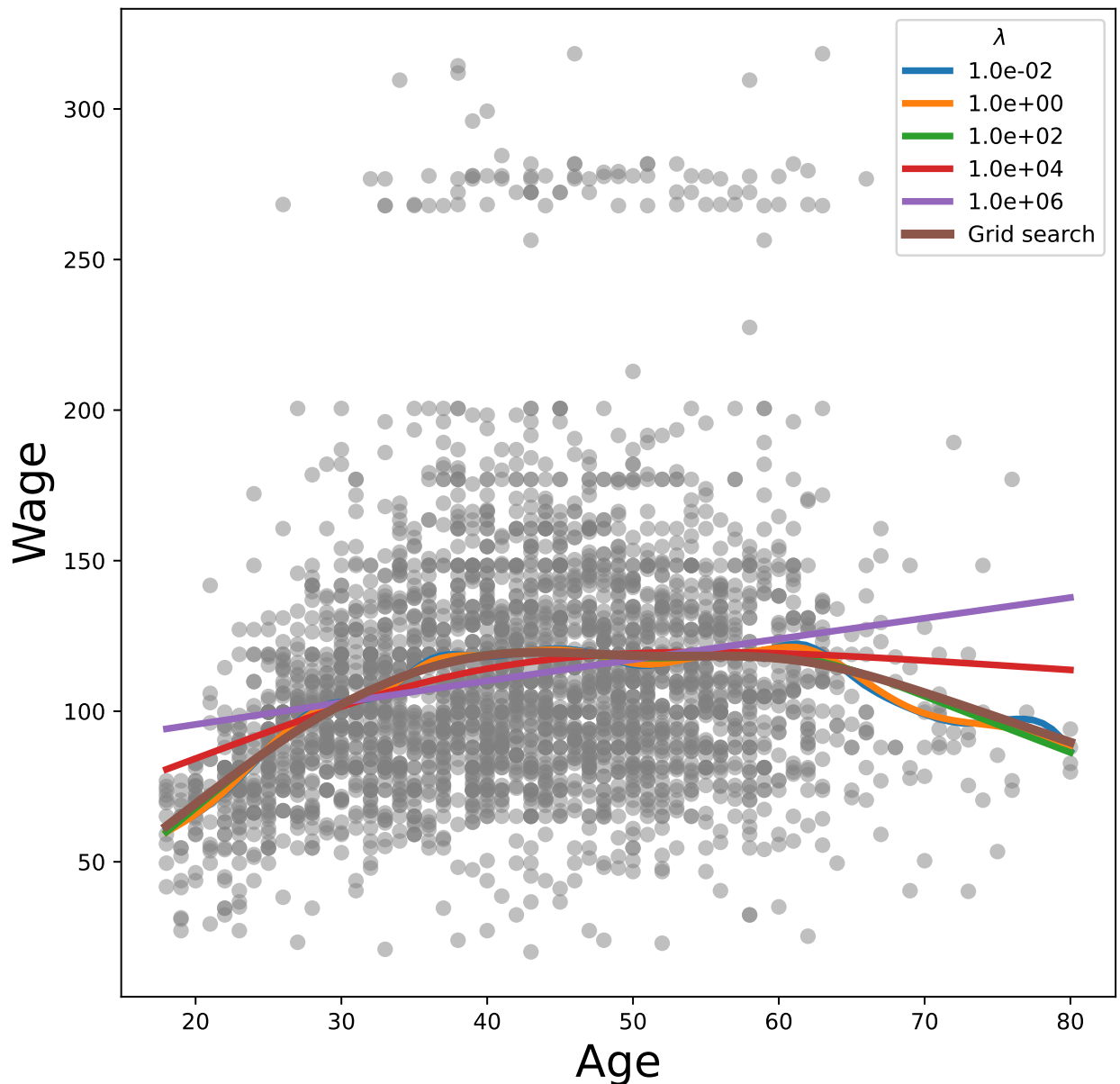


**Exercise 17.** `pygam` can search for an optimal smoothing parameter. Reproduce the plot from Exercise 16, call `.gridsearch(X_age, y)`, and add the resulting fit on top.

## Solution

```
fig, ax = subplots(figsize=(8, 8))
ax.scatter(age, y, facecolor='gray', alpha=0.5)
for lam in np.logspace(-2, 6, 5):
    gam = LinearGAM(s_gam(0, lam=lam)).fit(X_age, y)
    ax.plot(age_grid,
            gam.predict(age_grid),
            label='{:.1e}'.format(lam),
            linewidth=3)
gam_opt = gam.gridsearch(X_age, y)
ax.plot(age_grid,
        gam_opt.predict(age_grid),
        label='Grid search',
        linewidth=4)
ax.set_xlabel('Age', fontsize=20)
ax.set_ylabel('Wage', fontsize=20)
ax.legend(title=r'$\lambda$')
fig
```





**Exercise 18.** It's often more interpretable to specify the *effective degrees of freedom* than  $\lambda$ . Use `approx_lam()` from `ISLP.pygam` to find a  $\lambda$  giving roughly 4 degrees of freedom for the age term, and verify with `degrees_of_freedom()`.

Solution

```
age_term = gam.terms[0]
lam_4 = approx_lam(X_age, age_term, 4)
age_term.lam = lam_4
degrees_of_freedom(X_age, age_term)
```

```
np.float64(4.000000100000307)
```

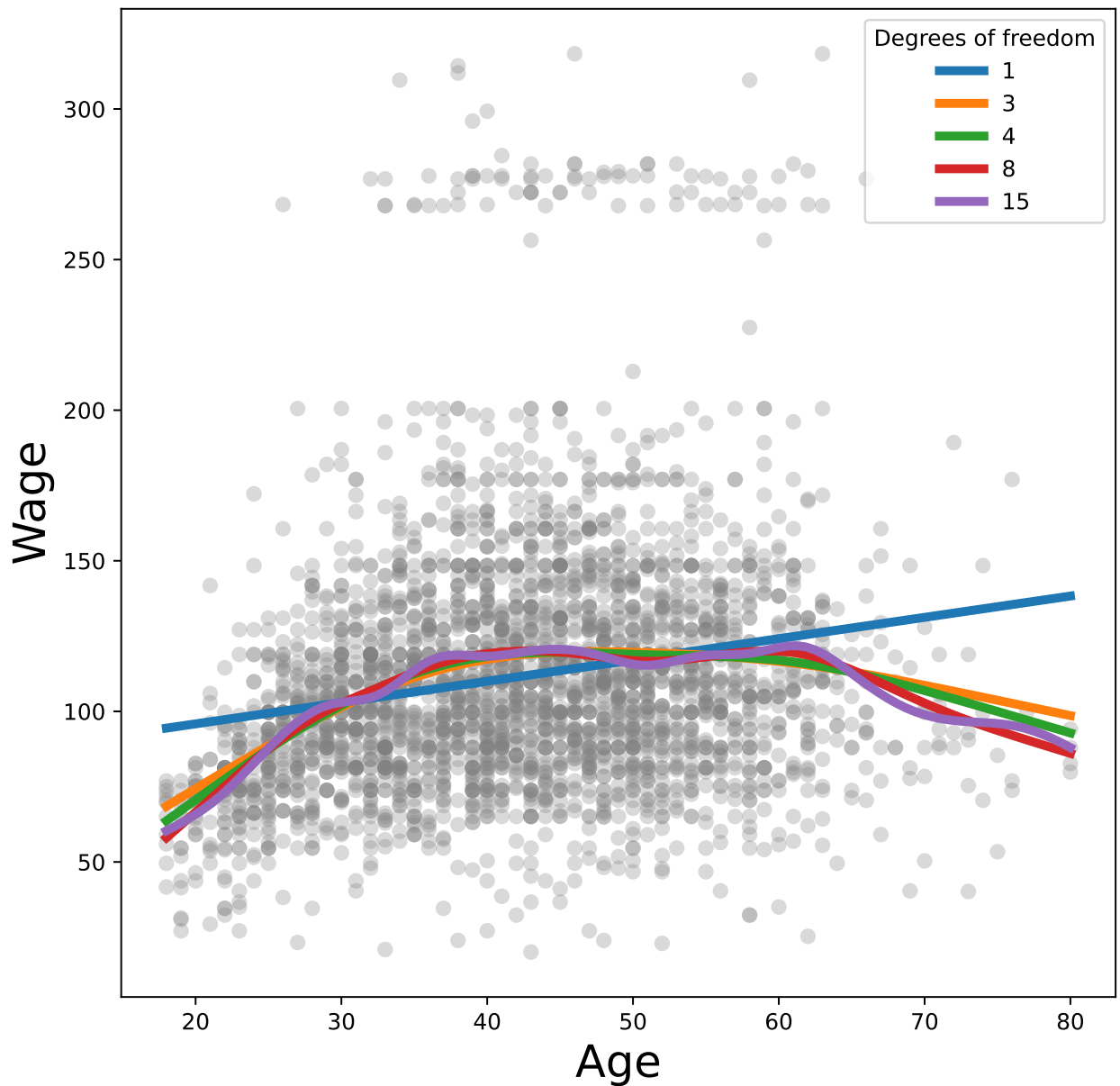
These degrees of freedom include the unpenalized intercept and linear term of the smoothing spline

— so there are at least 2 df.

**Exercise 19.** Now loop over df values [1, 3, 4, 8, 15] (passing df + 1 to `approx_lam()` to account for the intercept), refit, and plot each.

Solution

```
fig, ax = subplots(figsize=(8, 8))
ax.scatter(X_age,
           y,
           facecolor='gray',
           alpha=0.3)
for df in [1, 3, 4, 8, 15]:
    lam = approx_lam(X_age, age_term, df + 1)
    age_term.lam = lam
    gam.fit(X_age, y)
    ax.plot(age_grid,
            gam.predict(age_grid),
            label='{ :d}'.format(df),
            linewidth=4)
ax.set_xlabel('Age', fontsize=20)
ax.set_ylabel('Wage', fontsize=20);
ax.legend(title='Degrees of freedom');
```



A value of  $df=1$  corresponds to a straight-line fit.

#### 4.1 Additive Models with Several Terms

**Exercise 20.** Build a GAM “by hand”: use natural splines (`NaturalSpline`) with  $df=4$  for age and  $df=5$  for year, get dummies for education, stack them all into a model matrix  $X_{bh}$ , and fit OLS.

### Solution

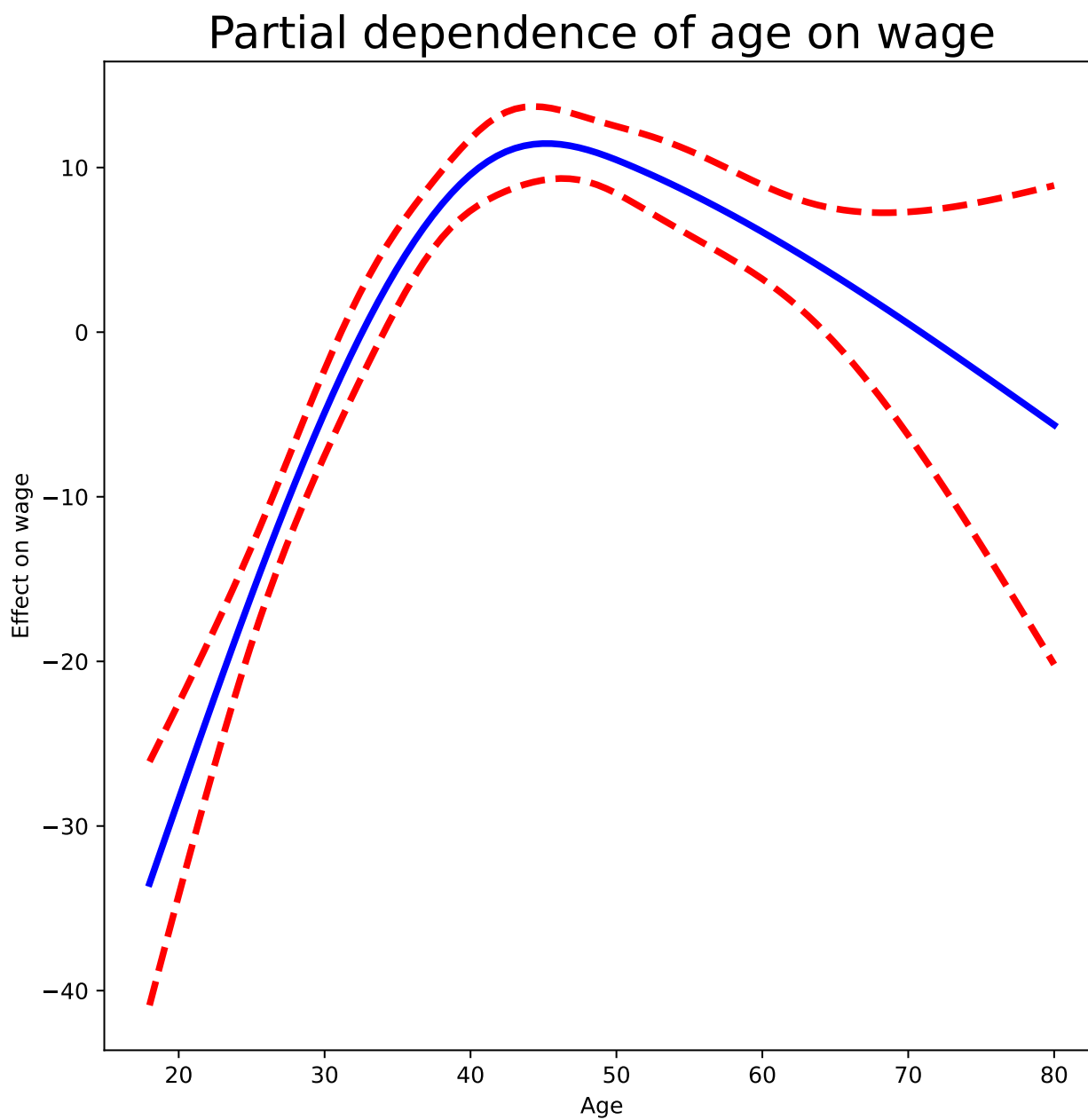
```
ns_age = NaturalSpline(df=4).fit(age)
ns_year = NaturalSpline(df=5).fit(Wage['year'])
Xs = [ns_age.transform(age),
      ns_year.transform(Wage['year']),
      pd.get_dummies(Wage['education']).values]
X_bh = np.hstack(Xs)
gam_bh = sm.OLS(y, X_bh).fit()
```

We use all columns of the indicator matrix for `education` (no intercept), and stack the three component matrices horizontally to form `X_bh`.

**Exercise 21.** Construct a partial-dependence plot for `age` from the by-hand GAM: predict on a matrix where all columns except those for `age` are held at their column means, and the `age` columns are swept across `age_grid`.

### Solution

```
age_grid = np.linspace(age.min(),
                       age.max(),
                       100)
X_age_bh = X_bh.copy()[1:100]
X_age_bh[:, 0] = X_bh.mean(0)[None, :]
X_age_bh[:, 4] = ns_age.transform(age_grid)
preds = gam_bh.get_prediction(X_age_bh)
bounds_age = preds.conf_int(alpha=0.05)
partial_age = preds.predicted_mean
center = partial_age.mean()
partial_age -= center
bounds_age -= center
fig, ax = subplots(figsize=(8, 8))
ax.plot(age_grid, partial_age, 'b', linewidth=3)
ax.plot(age_grid, bounds_age[:, 0], 'r--', linewidth=3)
ax.plot(age_grid, bounds_age[:, 1], 'r--', linewidth=3)
ax.set_xlabel('Age')
ax.set_ylabel('Effect on wage')
ax.set_title('Partial dependence of age on wage', fontsize=20);
```



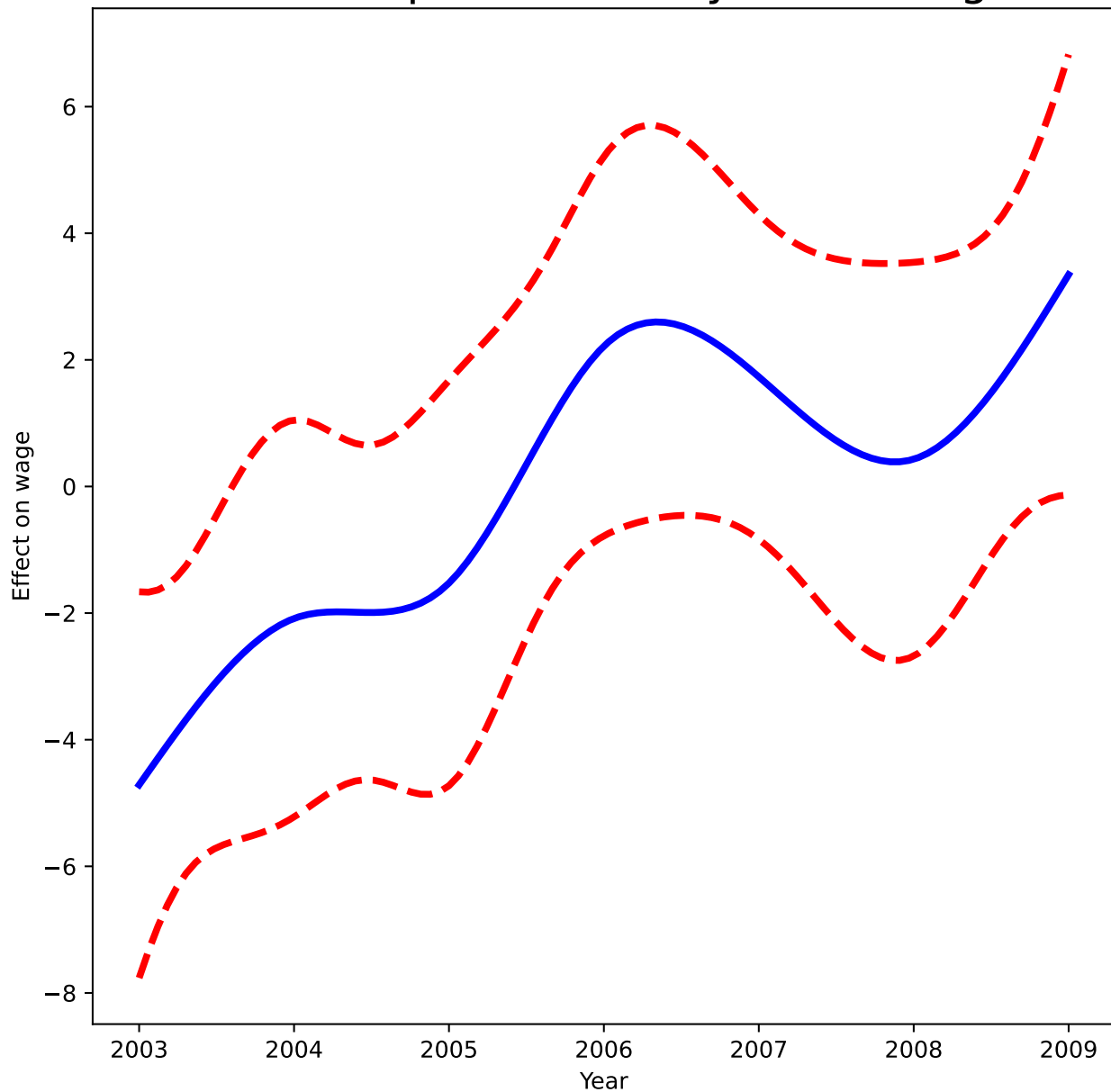
The idea is to create a new prediction matrix in which all but the columns for `age` are held constant at the training-data means; the four `age` columns are filled with the natural spline basis evaluated at the 100 values in `age_grid`.

**Exercise 22.** Do the same for `year`.

## Solution

```
year_grid = np.linspace(Wage['year'].min(),
                        Wage['year'].max(),
                        100)
X_year_bh = X_bh.copy()[100]
X_year_bh[:] = X_bh.mean(0)[None, :]
X_year_bh[:, 4:9] = ns_year.transform(year_grid)
preds = gam_bh.get_prediction(X_year_bh)
bounds_year = preds.conf_int(alpha=0.05)
partial_year = preds.predicted_mean
center = partial_year.mean()
partial_year -= center
bounds_year -= center
fig, ax = subplots(figsize=(8, 8))
ax.plot(year_grid, partial_year, 'b', linewidth=3)
ax.plot(year_grid, bounds_year[:, 0], 'r--', linewidth=3)
ax.plot(year_grid, bounds_year[:, 1], 'r--', linewidth=3)
ax.set_xlabel('Year')
ax.set_ylabel('Effect on wage')
ax.set_title('Partial dependence of year on wage', fontsize=20);
```

## Partial dependence of year on wage



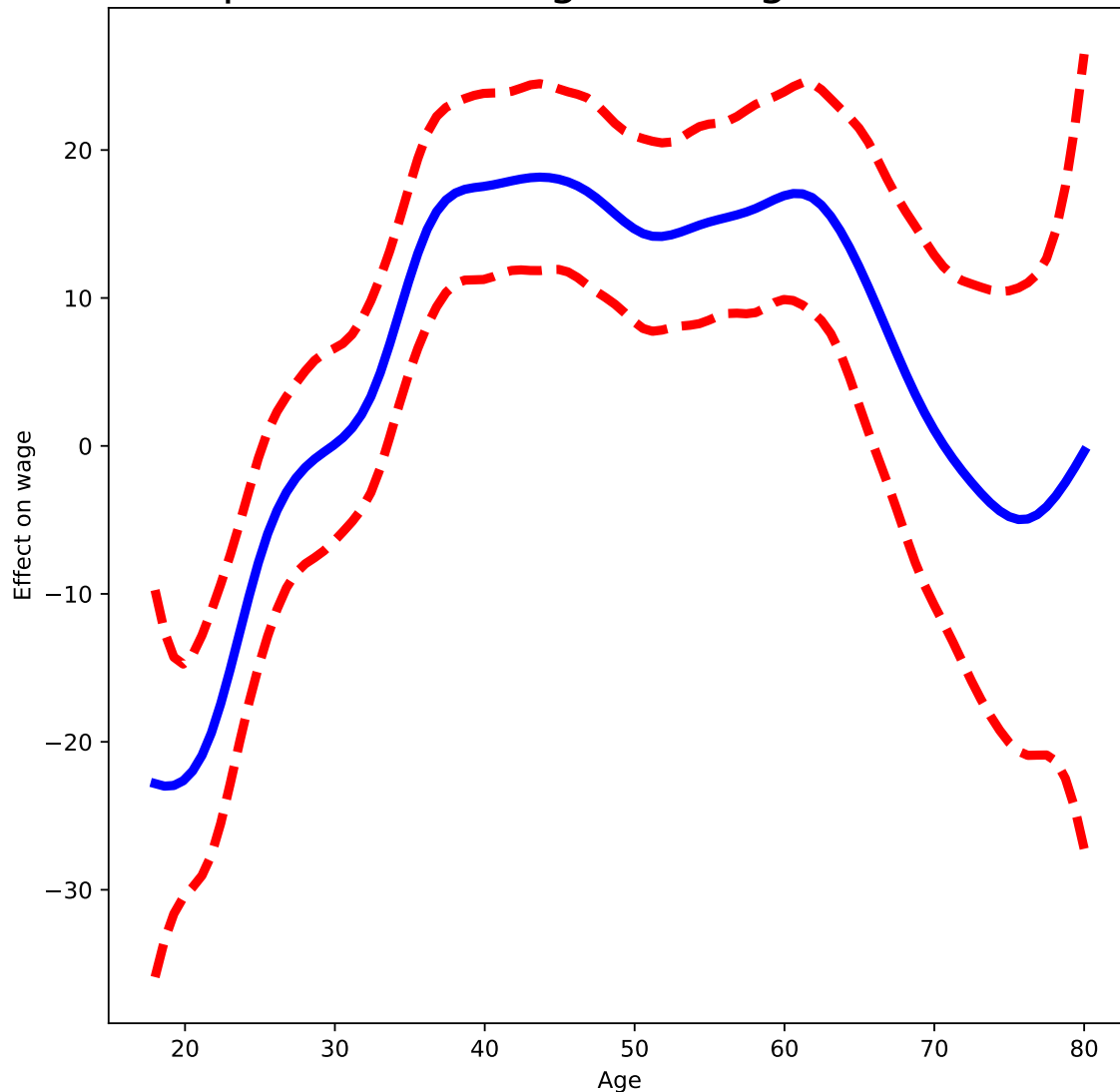
**Exercise 23.** Now fit the same GAM using *smoothing* splines via `pygam`. Build `LinearGAM(s_gam(0) + s_gam(1, n_splines=7) + f_gam(2, lam=0))`, where `f_gam` is a factor term for education. (Convert education to numeric codes with `.cat.codes`.) Then plot the partial dependence for age using `plot_gam()` from `ISLP.pygam`.

## Solution

```
gam_full = LinearGAM(s_gam(0) +  
                     s_gam(1, n_splines=7) +  
                     f_gam(2, lam=0))  
Xgam = np.column_stack([age,  
                        Wage['year'],  
                        Wage['education'].cat.codes])  
gam_full = gam_full.fit(Xgam, y)
```

```
fig, ax = subplots(figsize=(8, 8))  
plot_gam(gam_full, 0, ax=ax)  
ax.set_xlabel('Age')  
ax.set_ylabel('Effect on wage')  
ax.set_title('Partial dependence of age on wage - default lam=0.6', fontsize=20);
```

## Partial dependence of age on wage - default lam=0.6



`year` has only 7 unique values, so we use 7 basis functions. `lam=0` on the factor term avoids any

shrinkage. The default `lam=0.6` for `age` produces a noticeably wiggly fit.

**Exercise 24.** Re-tune `gam_full` so that the `age` and `year` smoothers each have ~4 degrees of freedom (use `approx_lam()` with `df=4+1`), then re-plot the partial dependence for `age` and for `year`.

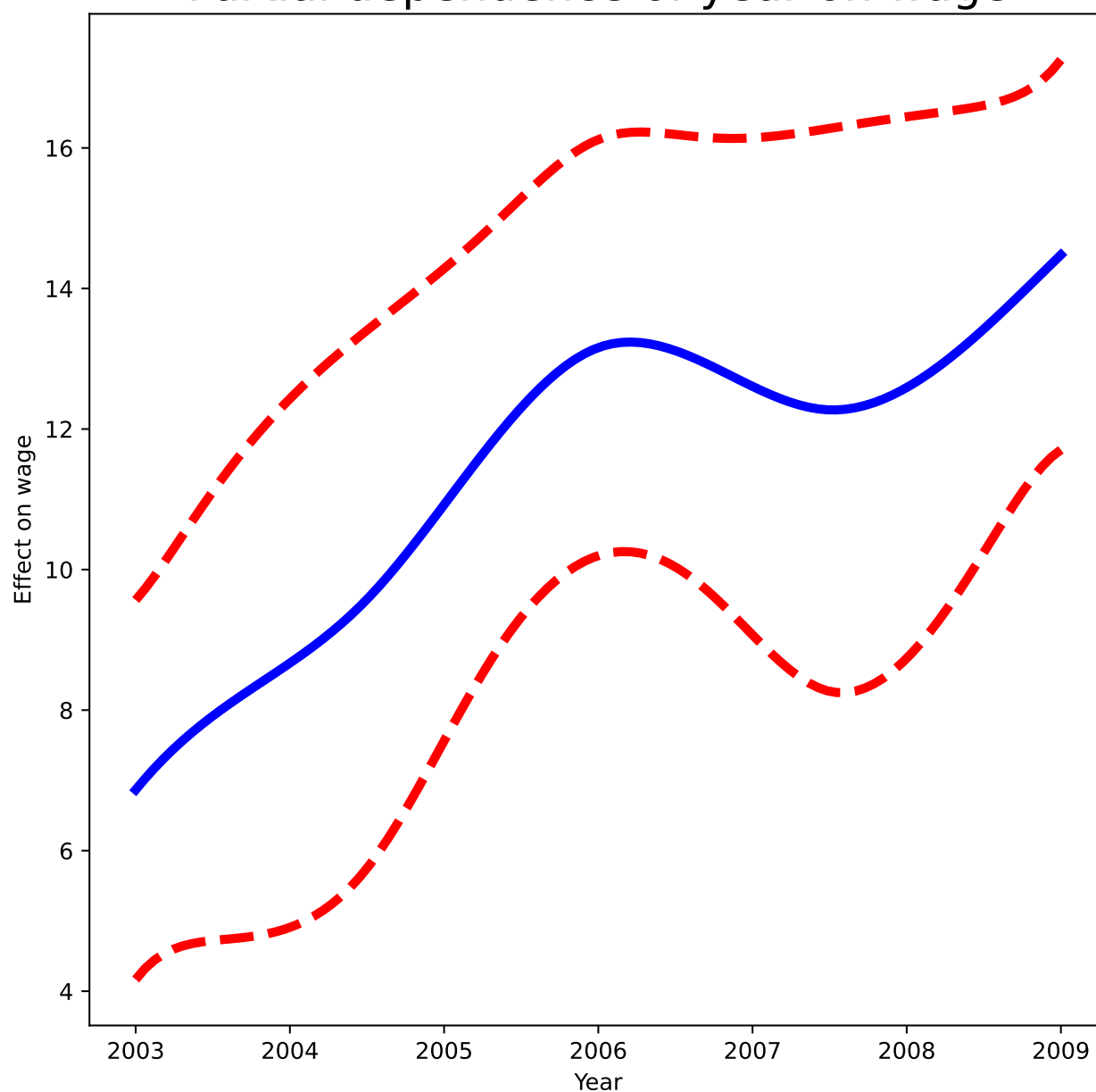
Solution

```
age_term = gam_full.terms[0]
age_term.lam = approx_lam(Xgam, age_term, df=4 + 1)
year_term = gam_full.terms[1]
year_term.lam = approx_lam(Xgam, year_term, df=4 + 1)
gam_full = gam_full.fit(Xgam, y)
```

Note that updating `age_term.lam` updates `gam_full.terms[0]` in place.

```
fig, ax = subplots(figsize=(8, 8))
plot_gam(gam_full,
         1,
         ax=ax)
ax.set_xlabel('Year')
ax.set_ylabel('Effect on wage')
ax.set_title('Partial dependence of year on wage', fontsize=20)
```

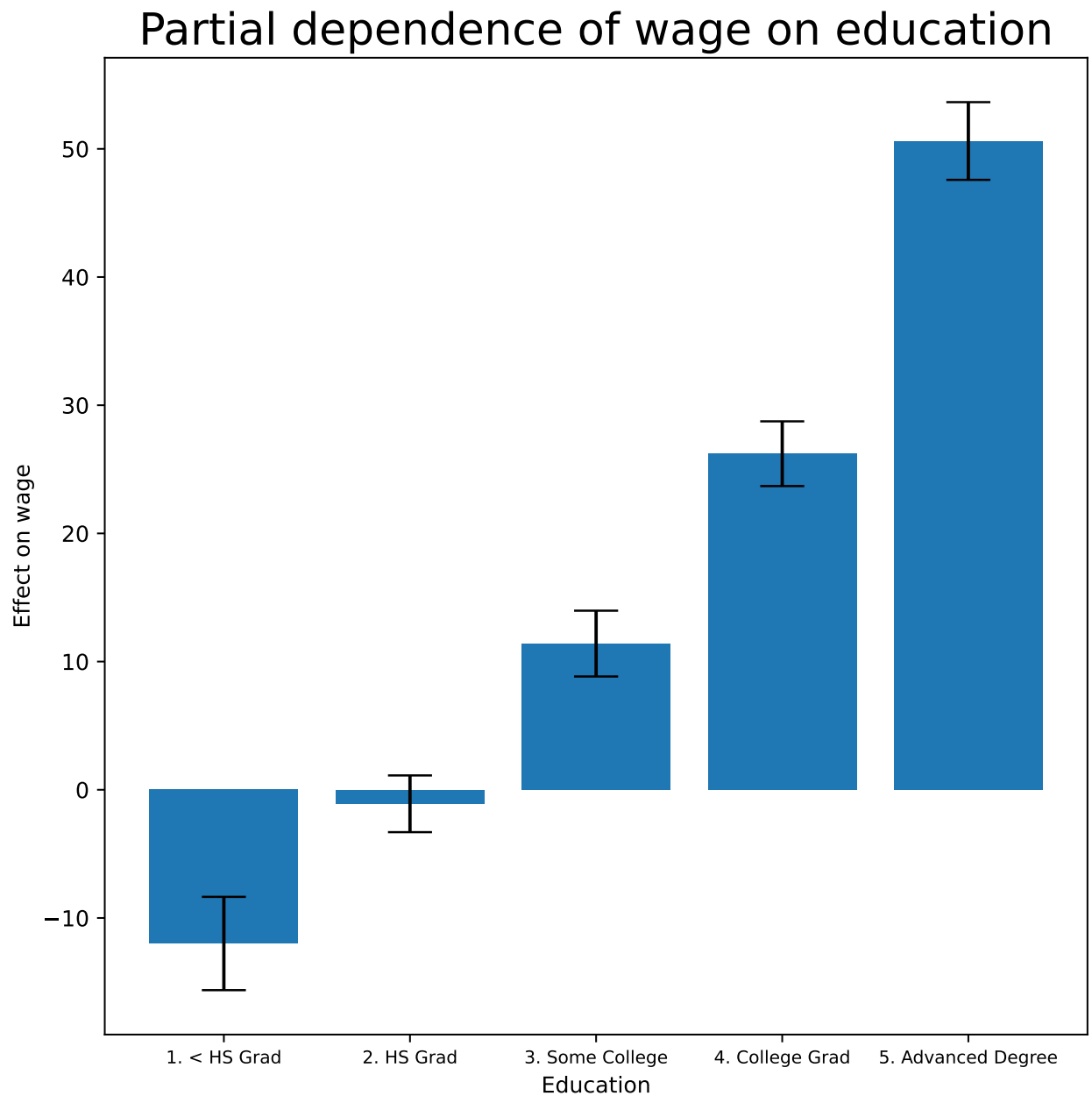
## Partial dependence of year on wage



**Exercise 25.** Plot the partial dependence for `education` — a categorical term gets a different style of plot, suited to a set of fitted constants per level.

## Solution

```
fig, ax = subplots(figsize=(8, 8))
ax = plot_gam(gam_full, 2)
ax.set_xlabel('Education')
ax.set_ylabel('Effect on wage')
ax.set_title('Partial dependence of wage on education',
             fontsize=20);
ax.set_xticklabels(Wage['education'].cat.categories, fontsize=8);
```



## 4.2 ANOVA Tests for Additive Models

**Exercise 26.** In all our models, the function of `year` looks fairly linear. Compare three models by ANOVA: a GAM that *excludes* `year` ( $\mathcal{M}_1$ ), a GAM that uses a *linear* term in `year` ( $\mathcal{M}_2$ ), and the full GAM with a smoothing spline in `year` ( $\mathcal{M}_3$ ). What do you conclude?

Solution

```
gam_0 = LinearGAM(age_term + f_gam(2, lam=0))
gam_0.fit(Xgam, y)
gam_linear = LinearGAM(age_term +
                        l_gam(1, lam=0) +
                        f_gam(2, lam=0))
gam_linear.fit(Xgam, y)

anova_gam(gam_0, gam_linear, gam_full)
```

|   | deviance      | df          | deviance_diff | df_diff  | F        | pvalue   |
|---|---------------|-------------|---------------|----------|----------|----------|
| 0 | 105402.432206 | 2991.004005 | NaN           | NaN      | NaN      | NaN      |
| 1 | 105134.624170 | 2990.005190 | 267.808037    | 0.998815 | 7.625329 | 0.016210 |
| 2 | 105030.684668 | 2987.007254 | 103.939502    | 2.997936 | 0.986003 | 0.429637 |

Note the reuse of `age_term` — that term already has its `lam` set to yield 4 df, and we want to keep that across all three models.

There is compelling evidence that a GAM with a linear `year` term is better than one that omits `year` ( $p = 0.002$ ). However, there is no evidence that a nonlinear function of `year` is needed ( $p = 0.435$ ). So  $\mathcal{M}_2$  is preferred.

**Exercise 27.** Repeat the test for `age`.

Solution

```
gam_0 = LinearGAM(year_term +
                  f_gam(2, lam=0))
gam_linear = LinearGAM(l_gam(0, lam=0) +
                       year_term +
                       f_gam(2, lam=0))
gam_0.fit(Xgam, y)
gam_linear.fit(Xgam, y)
anova_gam(gam_0, gam_linear, gam_full)
```

|   | deviance      | df          | deviance_diff | df_diff  | F         | pvalue   |
|---|---------------|-------------|---------------|----------|-----------|----------|
| 0 | 109043.809955 | 2991.000589 | NaN           | NaN      | NaN       | NaN      |
| 1 | 107295.111571 | 2990.000704 | 1748.698385   | 0.999884 | 49.737638 | 0.000009 |
| 2 | 105030.684668 | 2987.007254 | 2264.426903   | 2.993450 | 21.513266 | 0.000026 |

Very strong evidence that a nonlinear term is required for `age`.

**Exercise 28.** Print `gam_full.summary()` and use `gam_full.predict()` to generate fitted values on the training set.

Solution

```
gam_full.summary()
```

LinearGAM

```
=====
Distribution:                NormalDist Effective DoF:
Link Function:              IdentityLink Log Likelihood:
14930.261
Number of Samples:          3000 AIC:
                               AICc:
                               GCV:
                               Scale:
                               Pseudo R-Squared:
=====
```

| Feature   | Function | Lambda     | Rank | EDoF | P > x  | S |
|-----------|----------|------------|------|------|--------|---|
| s(0)      |          | [465.0491] | 20   | 5.1  | 1.11e- |   |
| 16        | ***      |            |      |      |        |   |
| s(1)      |          | [2.1564]   | 7    | 4.0  | 8.10e- |   |
| 03        | **       |            |      |      |        |   |
| f(2)      |          | [0]        | 5    | 4.0  | 1.11e- |   |
| 16        | ***      |            |      |      |        |   |
| intercept |          |            | 1    | 0.0  | 1.11e- |   |
| 16        | ***      |            |      |      |        |   |

```
=====
Significance codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

WARNING: Fitting splines and a linear function to a feature introduces a model identifiability which can cause p-values to appear significant when they are not.

WARNING: p-values calculated in this manner behave correctly for un-penalized models or models with known smoothing parameters, but when smoothing parameters have been estimated, the p-values are typically lower than they should be, meaning that the tests reject the null too re

```
Yhat = gam_full.predict(Xgam)
```

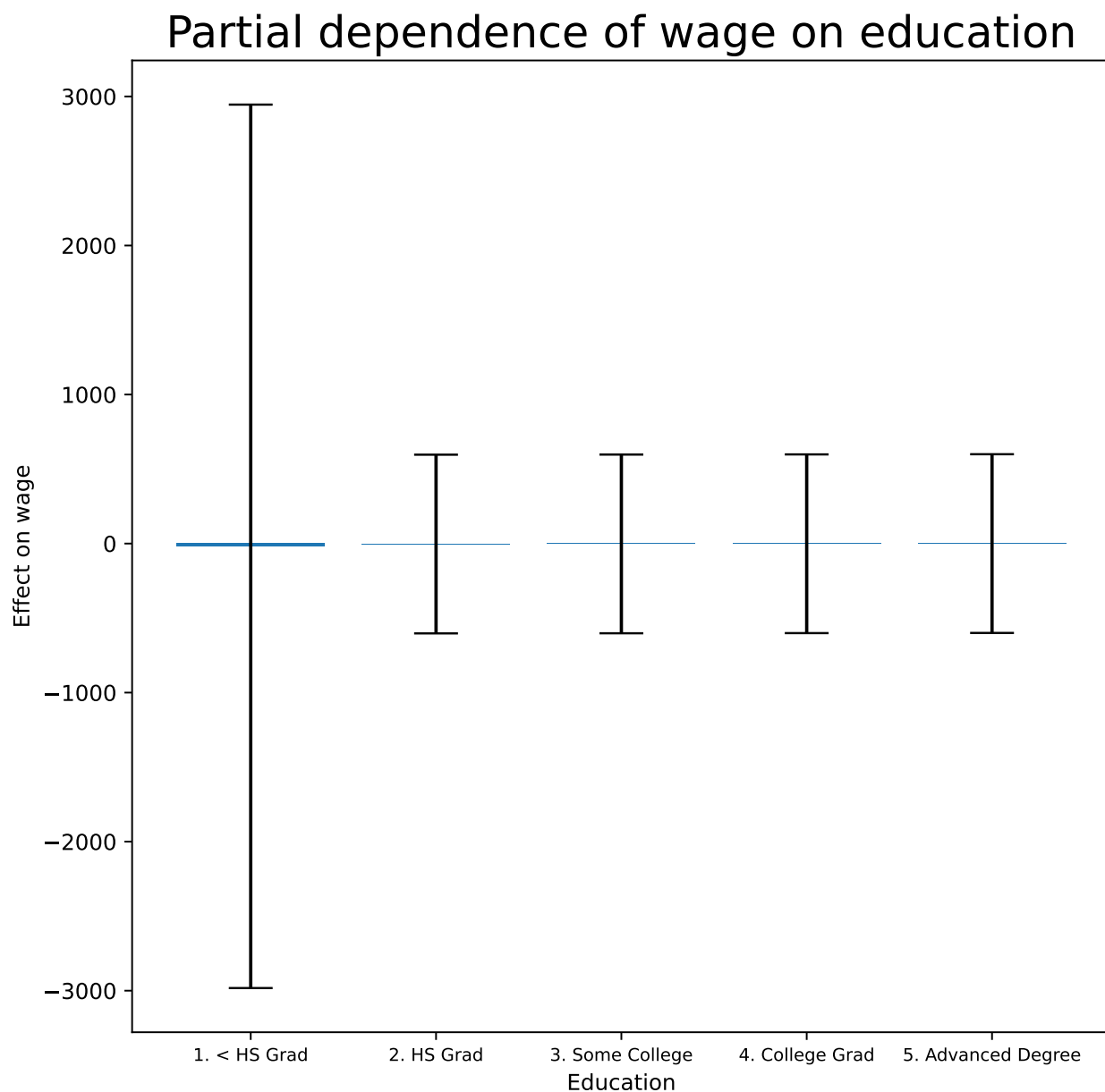
**Exercise 29.** Fit a logistic GAM for `high_earn` (i.e. `wage > 250`) using `LogisticGAM(age_term + l_gam(1, lam=0) + f_gam(2, lam=0))`. Plot the partial dependence for education.

## Solution

```
gam_logit = LogisticGAM(age_term +
                        l_gam(1, lam=0) +
                        f_gam(2, lam=0))
gam_logit.fit(Xgam, high_earn)

LogisticGAM(callbacks=[Deviance(), Diffs(), Accuracy()],
             fit_intercept=True, max_iter=100,
             terms=s(0) + l(1) + f(2) + intercept, tol=0.0001, verbose=False)

fig, ax = subplots(figsize=(8, 8))
ax = plot_gam(gam_logit, 2)
ax.set_xlabel('Education')
ax.set_ylabel('Effect on wage')
ax.set_title('Partial dependence of wage on education',
             fontsize=20);
ax.set_xticklabels(Wage['education'].cat.categories, fontsize=8);
```



The fit looks flat with very wide error bars on the first category.

**Exercise 30.** Cross-tabulate `high_earn` against `education`. What do you see?

Solution

```
pd.crosstab(Wage['high_earn'], Wage['education'])
```

|           | 1. < HS Grad | 2. HS Grad | 3. Some College | 4. College Grad | 5. Advanced Degree |
|-----------|--------------|------------|-----------------|-----------------|--------------------|
| education |              |            |                 |                 |                    |
| high_earn |              |            |                 |                 |                    |
| False     | 268          | 966        | 643             | 663             | 381                |
| True      | 0            | 5          | 7               | 22              | 45                 |

There are *no* high earners in the lowest education category, so the logistic GAM has nothing to estimate there.

**Exercise 31.** Refit the logistic GAM excluding observations in the “1. < HS Grad” category, then plot the partial dependence for `education`, `year`, and `age`.

Solution

```
only_hs = Wage['education'] == '1. < HS Grad'
Wage_ = Wage.loc[~only_hs]
Xgam_ = np.column_stack([Wage_['age'],
                        Wage_['year'],
                        Wage_['education'].cat.codes - 1])
high_earn_ = Wage_['high_earn']
```

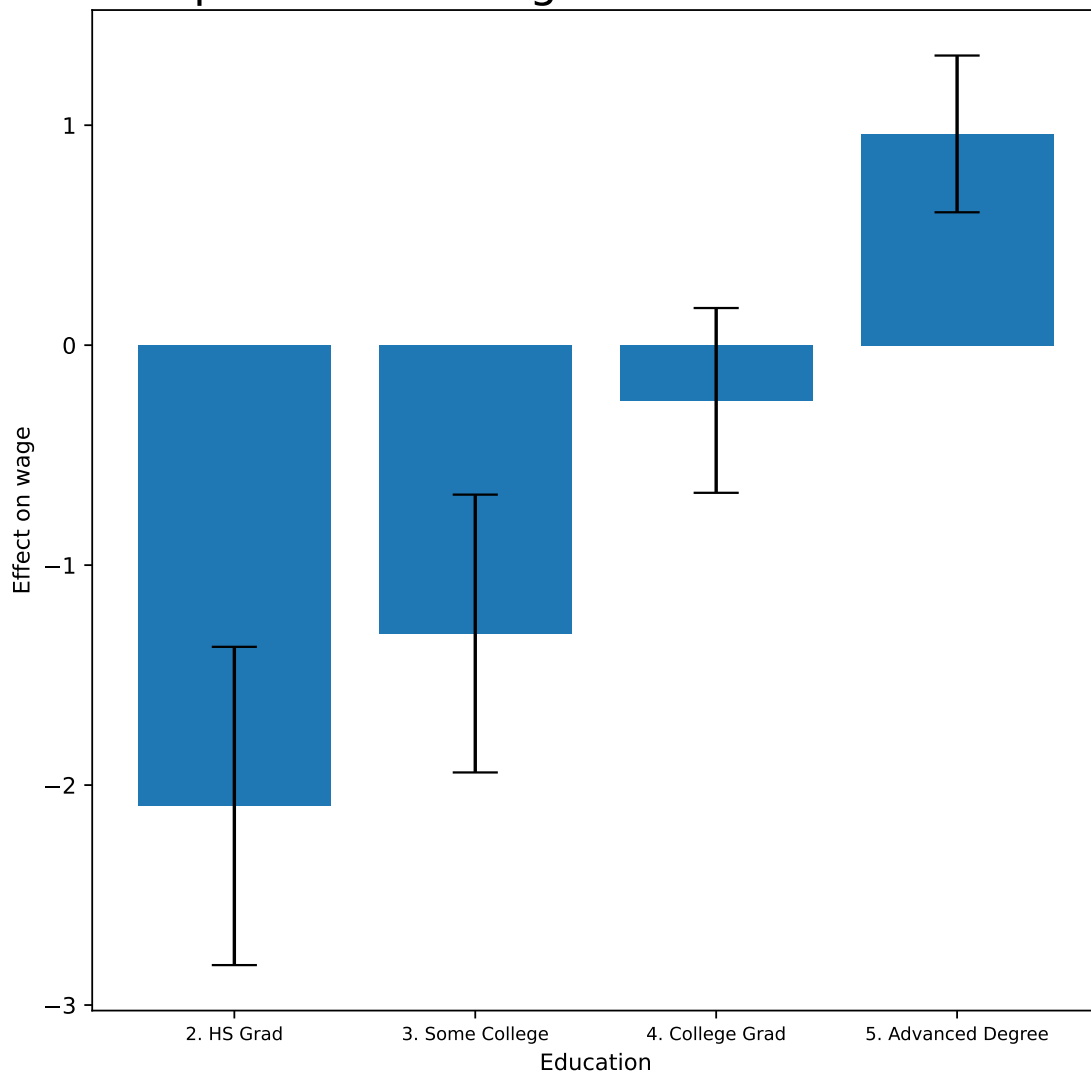
We subtract 1 from `cat.codes` to work around a `pygam` bug — the relabeling has no effect on the fit.

```
gam_logit_ = LogisticGAM(age_term +
                        year_term +
                        f_gam(2, lam=0))
gam_logit_.fit(Xgam_, high_earn_)
```

```
LogisticGAM(callbacks=[Deviance(), Diffs(), Accuracy()],
            fit_intercept=True, max_iter=100,
            terms=s(0) + s(1) + f(2) + intercept, tol=0.0001, verbose=False)
```

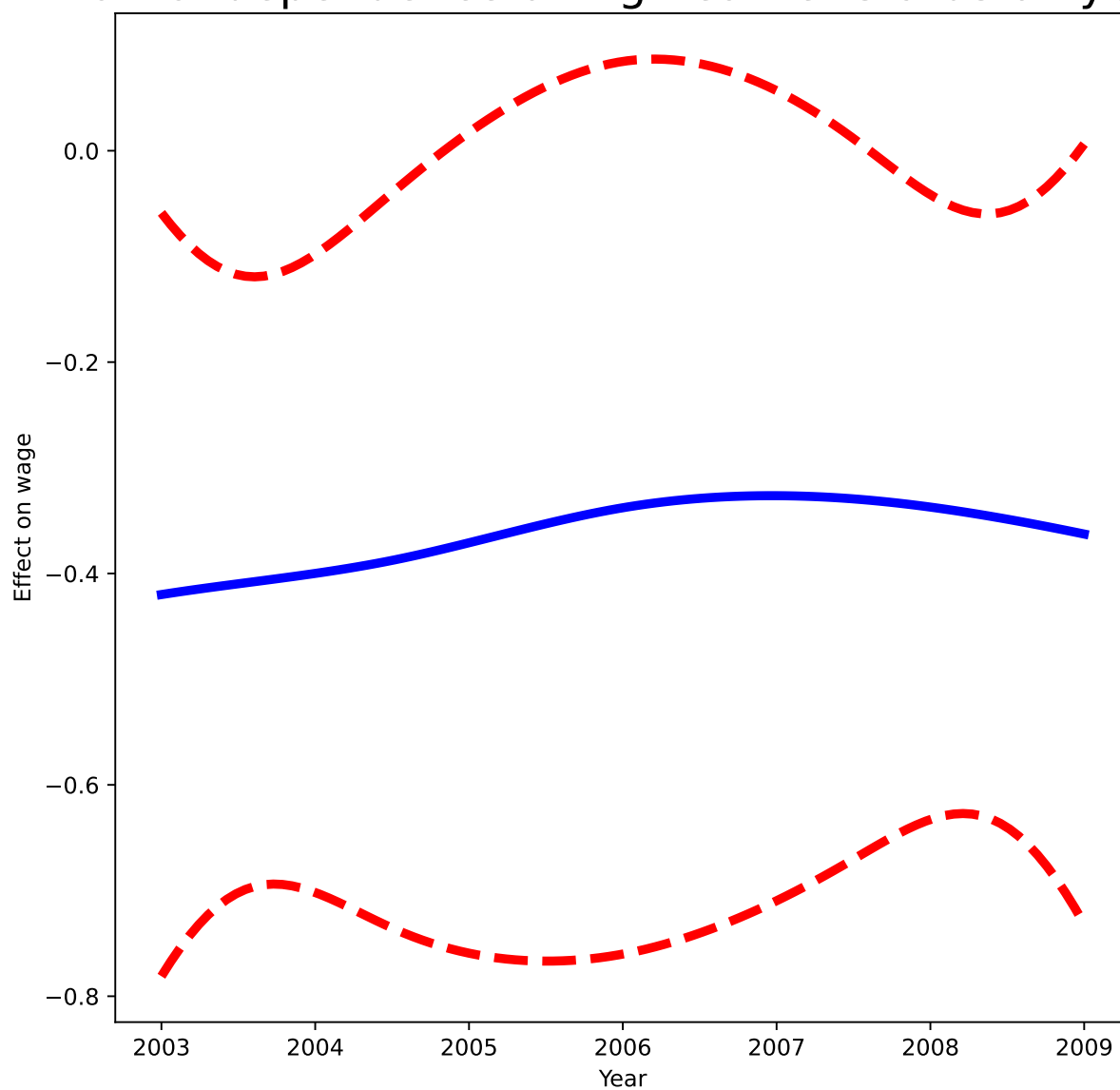
```
fig, ax = subplots(figsize=(8, 8))
ax = plot_gam(gam_logit_, 2)
ax.set_xlabel('Education')
ax.set_ylabel('Effect on wage')
ax.set_title('Partial dependence of high earner status on education', fontsize=20);
ax.set_xticklabels(Wage['education'].cat.categories[1:],
                  fontsize=8);
```

## Partial dependence of high earner status on education



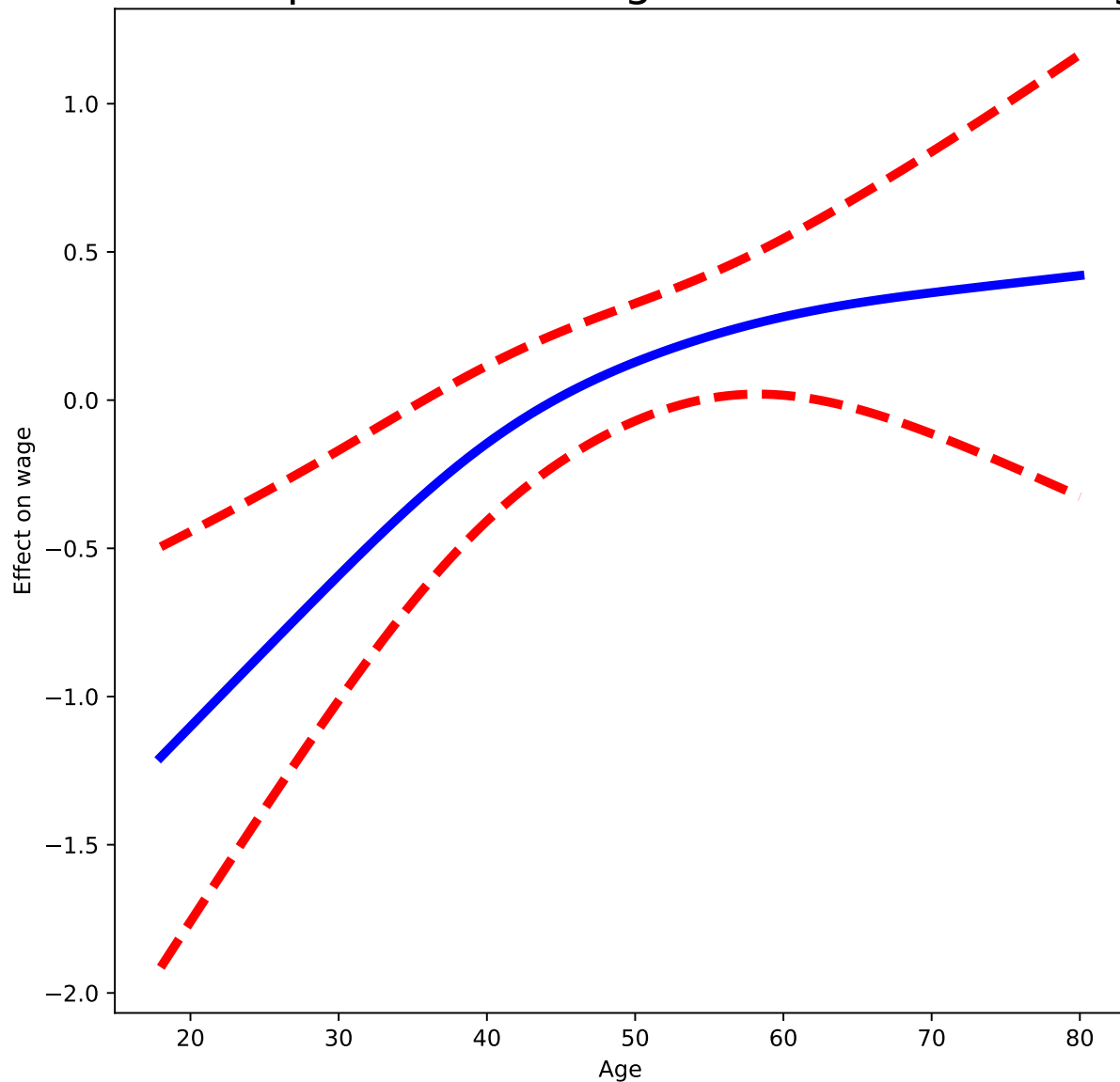
```
fig, ax = subplots(figsize=(8, 8))
ax = plot_gam(gam_logit_, 1)
ax.set_xlabel('Year')
ax.set_ylabel('Effect on wage')
ax.set_title('Partial dependence of high earner status on year',
             fontsize=20);
```

## Partial dependence of high earner status on year



```
fig, ax = subplots(figsize=(8, 8))
ax = plot_gam(gam_logit_, 0)
ax.set_xlabel('Age')
ax.set_ylabel('Effect on wage')
ax.set_title('Partial dependence of high earner status on age', fontsize=20);
```

## Partial dependence of high earner status on age

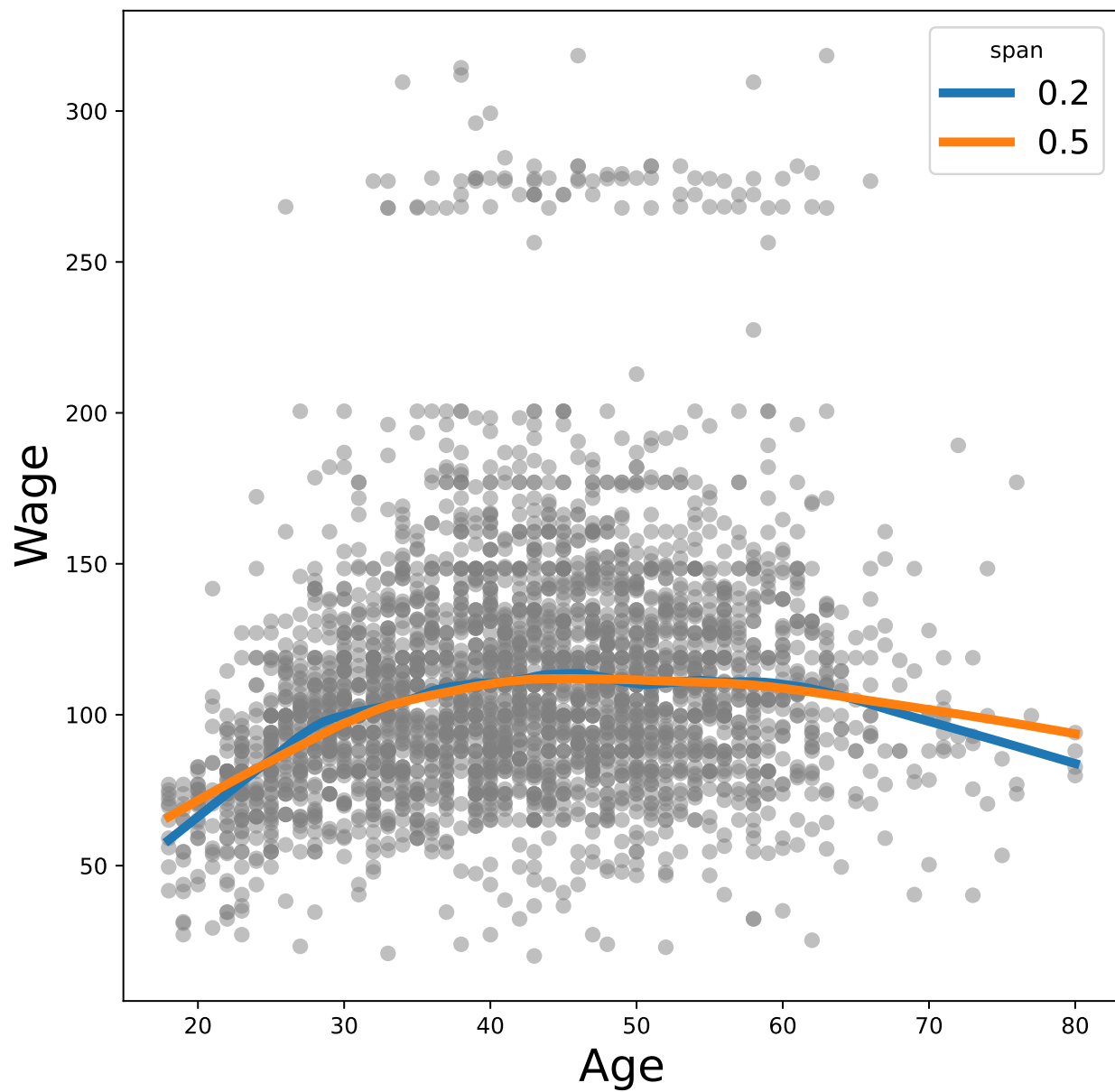


## 5 Local Regression

**Exercise 32.** Finally, use `sm.nonparametric.lowess` to fit local linear regression models with spans of 0.2 and 0.5, and plot them together. Some implementations of GAMs allow local-regression terms; `pygam` does not.

## Solution

```
lowess = sm.nonparametric.lowess
fig, ax = subplots(figsize=(8, 8))
ax.scatter(age, y, facecolor='gray', alpha=0.5)
for span in [0.2, 0.5]:
    fitted = lowess(y,
                    age,
                    frac=span,
                    xvals=age_grid)
    ax.plot(age_grid,
            fitted,
            label='{:.1f}'.format(span),
            linewidth=4)
ax.set_xlabel('Age', fontsize=20)
ax.set_ylabel('Wage', fontsize=20);
ax.legend(title='span', fontsize=15);
```



As expected, a span of 0.5 produces a smoother fit than 0.2.